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Bridging Atomistic Simulations and Reservoir Computing for Predicting Structural and Transport Properties of Thiol−Ene Click-Cross-Linked Carboxymethyl Cellulose Hydrogels

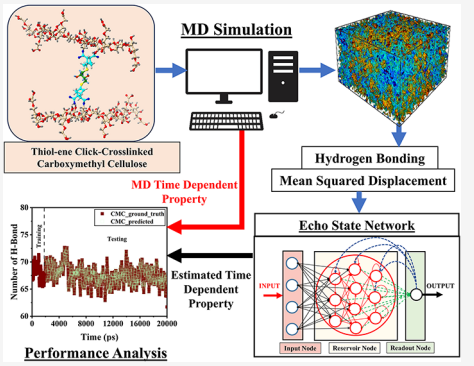
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ABSTRACT: The long-time scale behavior of hydrogels is still a core problem in material design, especially due to the limitation of computational cost and accessibility time scale of molecular dynamics (MD) simulations. So, this study offered a combined model, which integrates both MD and artificial intelligence (AI) models to truly predict the dynamics of structural and transport characteristics of thiol-norbornene click cross-linked carboxymethyl cellulose (CMC) hydrogels. All-atom MD portrays a consistent network of hydrogen bonds, a measurable swelling ratio, and restricted polymer mobility, and these are indicators of a strong gel. To extrapolate these properties, the gated recurrent unit (GRU), long short-term memory (LSTM), and transformer-based LAG-LLAMA are trained and compared with an echo state network (ESN). The GRU, LSTM, and LAG-LLAMA models have not been successful in explaining the nonlinear oscillations of hydrogen bonding, obtaining insignificant values of R^2 (0.05), whereas ESN has demonstrated outstanding predictive power (R^2 = 0.99) with only 200 MD trajectory frames of initial information. In addition to this, despite the deep learning models achieving a high accuracy, via optimization, for mean-squared displacement (R^2 = 0.95), the training sets are much larger (40−70%) than the ESN. In this way, the present work shows that reservoir computing models are more efficient and reachable in data efficiency and stability compared to the traditional recurrent and transformer models when applied to modeling the time-dependent transformation of complex molecular systems. This methodology will provide a generalizable methodology of fastening the computational design of soft materials bridging atomistic fidelity and AI-based temporal extrapolation.



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1. INTRODUCTION

Artificial intelligence (AI) plays a growing role in hydrogel property predictions by offering computational tools to understand and forecast complex material behaviors.[1−3] In materials science, AI supports structure−property mapping,[4,5] phase prediction,[6] and optimization of synthesis conditions,[7] reducing the need for extensive experimental trials. Carboxymethyl cellulose (CMC)-based hydrogels are an important focus due to their biocompatibility, adjustable mechanical and chemical features, and applications in drug delivery, tissue engineering, and wound healing, making them strong candidates for AI-driven modeling.[8−10] Neural networks, a subset of AI, are particularly effective for property prediction because they can capture higher orders of nonlinear relationships between atomic structures and bulk properties.[11,12] Various architectures serve different roles: feedforward networks handle static predictions, convolutional neural networks (CNNs) process molecular images, recurrent neural networks (RNNs) address sequential data, and graph neural networks (GNNs) model molecular connectivity.[11−13] However, time-dependent data in materials research presents challenges, as many networks struggle to capture long-term dependencies.[14,15] Long short-term memory (LSTM) networks can model extended sequences but are computationally heavy, while gated recurrent units (GRUs) train faster but may lose accuracy over very long sequences.[16] Transformer-based models excel at complex temporal patterns through attention mechanisms but require large data sets and high computational resources.[16] Liquid neural networks improve efficiency and maintain long-term memory using adaptive time constants. A specific type of liquid neural network is known as the echo state networks (ESNs). These networks provide a simpler and more computationally efficient approach to modeling temporal dynamics. By utilizing a fixed recurrent reservoir and only training the output layer, ESNs effectively reduce the risk of overfitting.[17,18] Despite these advances, acquiring high-quality time-dependent experimental data remains difficult due to limitations in real-time measurement techniques and the high cost of long-term monitoring.

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To address the challenge of obtaining time-dependent material data, the molecular dynamics (MD) simulation offers a computational approach to observe molecular behavior at the atomic scale. MD provides access to key structural and transport properties such as diffusivity, ionic conductivity, viscosity, hydrogen bonding (Hbond) patterns, and conformational changes, which are essential for predicting material evolution over time.[19−23] Studies using MD have examined crystalline cellulose surface interactions in aqueous environments, revealing hydrophilicity variations, Hbond networks, and conformational differences between amorphous and crystalline phases, along with glass transition temperature deviations from experiments.[24,25] Further work on oligomeric cellulose derivatives in ionic liquids quantified density, solubility parameters, and polymerization effects,[26] while sodium-based CMC salt studies show enhanced salt tolerance with higher substitution degrees, and sodium chloride effects on viscosity and structure.[27] Most MD studies focus on crystalline or physically cross-linked CMC hydrogels, which are environmentally friendly and easy to prepare but suffer from inconsistent gelation, lower mechanical strength, and limited stability compared to chemical cross-linking.[28,29] Among chemical cross-linking strategies, such as epichlorohydrin, aldehydes, urea derivatives, and photo-cross-linking,[30,31] photoinduced thiol−ene chemistry offers rapid, cytocompatible gelation, tunable mechanics, and improved aqueous stability.[32−34] MD studies on thiol−ene cross-linked poly(vinyl alcohol) and polyethylene glycol show higher stability, greater Young's modulus, and stronger Hbonding than physically cross-linked systems.[20,21] While MD captures real-time dynamics with atomic resolution, it remains computationally expensive.

Integrating MD with AI can reduce cost by accelerating force field predictions, optimizing sampling, extracting kinetic information, and generating molecular structures.[35,36] The deep learning framework is used for predicting hydrogel's mechanical properties from mesoscopic network structures. Two models—a deep neural network (DNN) with handcrafted features and a 3D CNN incorporating structural and physical information—are trained on stress−stretch curves from 2200 simulated networks. Results show that while the DNN performs well for dense networks, it struggles with sparse ones, whereas the 3D CNN demonstrates superior accuracy and robustness by extracting spatial features directly.[37] Water diffusion in complex polymer systems, including copolymers, can be predicted with high accuracy and low processing cost by using an architecture based on Wasserstein generative adversarial neural networks. The model's latent variable distribution correlates with accuracy, enabling practical use in materials development by guiding input data length and reliability assessment.[38] The Langevin field-theoretic simulations (L-FTS) are accelerated by using a deep learning model by directly predicting saddle-point pressure fields. Using atrous convolutions and semantic segmentation architectures, the model achieves over six time speedup with high accuracy, robustness, and reusability, enabling efficient polymer simulation via open-source tools.[39] The charge-recursive neural network architecture is used to develop the ML-based force field calculation for liquid electrolytes of Li-ion batteries. The model accurately predicts key thermodynamic and transport properties, such as viscosity and diffusivity, without empirical fitting. It incorporates long-range interactions and charge equilibration, offering significant computational speedup over traditional density functional theory (DFT) methods. The model performs well across seven common carbonate solvents and shows strong agreement with experimental data.[40] Looking into the AI integrated with MD literature, major works have been done on predicting specific properties and force field developments. However, the AI framework is not used to predict the future materials' dynamic properties according to our knowledge.

In order to overcome this limitation, the main aim of this work is to construct a framework using MD simulation coupled with an ESN. The novelty of the current research is to use the ESN to understand complex time-dependent patterns based on short-size MD simulations in order to predict future structural and transport properties (e.g., Hbonding, mean-squared displacement (MSD)) with an appropriate accuracy and with low computational cost. This work presents a new approach for studying the long-term behavior of chemically cross-linked hydrogels, without having to use costly and time-consuming simulations. For this purpose, the work used a thiol−ene click cross-linked CMC hydrogel. These models are then immersed in excess water to mimic realistic hydrogel conditions, and MD simulations are performed to capture the molecular behavior at the atomic scale. From the simulation trajectories, key structural and transport properties such as Hbonding patterns, MSD, diffusivity, and viscosity are systematically calculated, as these parameters directly reflect the dynamic nature and stability of the hydrogel network. The extracted properties are then used as inputs for reservoir computing, which is a specific form of a liquid state neural network designed to handle time-dependent data efficiently. By training the model with this information, the network can learn the underlying temporal correlations and subsequently predict future material properties that extend beyond the time scales directly accessible through MD alone. This integrated MD-AI framework thus provides a computationally efficient approach for modeling and forecasting the behavior of thiol−ene click CMC hydrogels.

2. COMPUTATION METHODOLOGY

Figure 1 shows the flow diagram of this work, which begins with the creation of a thiol−ene click CMC polymer model. Once the model is prepared, a suitable force field is developed to capture the essential interactions. The polymer is then immersed in excess water to simulate realistic conditions, and MD simulations are carried out to observe the structural and

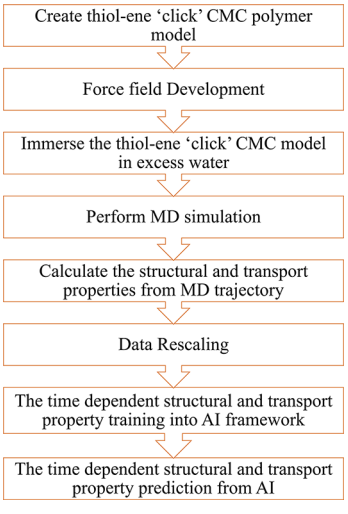


Figure 1. Flow diagram for this study.

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dynamic behavior of the system over time. From the MD trajectory, both the structural and transport properties are calculated to provide a detailed understanding of the system at the molecular scale. Since raw simulation data often vary in scale, a data rescaling step is applied to normalize the results for further use. The rescaled time-dependent structural and transport properties are then introduced as input for training an AI framework, which is designed to handle temporal patterns effectively. The AI framework learns from these data and generates predictions of time-dependent properties, extending beyond the MD simulations.

2.1. Polymer Building

Avogadro package[41] is used to draw the atomic structure of the CMC monomer. The CMC monomer is replicated five times to create an oligomeric CMC chain through an in-house program. The third CMC monomer in the oligomeric chain is functionalized with the norbornene molecule to enhance the material properties and prevent steric hindrance between the molecules. Further, the thiol moieties are introduced to the "ene" functional norbornene moiety, producing CMC functionalized with thiol−ene. Figure 2 shows three representative structures. Further, the CMC, norbornene-functionalized CMC (CMC-N), and thiol norbornene-functionalized CMC (CMC-NT) are replicated four times using an in-house Python program. So, the CMC, CMC-N, and CMC-NT have 20 repeating units each, as shown in Figure S1 of the Supporting Information (SI). However, the dithiothreitol functions as a crosslinker, as evidenced experimentally, connecting the norbornene-modified framework to create a gel framework. Consequently, the CMC-N and CMC-NT chains combine, forming a thiol−ene cross-linked CMC gel (CMC-NTC), as illustrated in Figure 2d. Several interconnected cross-linked gels create a framework, depicted in Figure S2 of the Supporting Information.

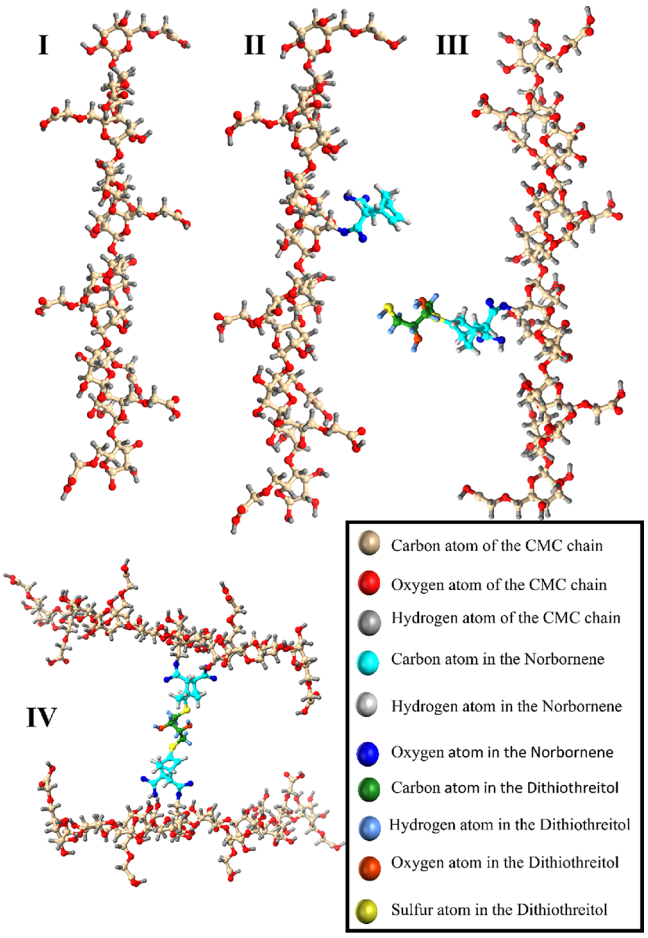


Figure 2. Initial configuration of polymer: (a) CMC, (b) CMC-N, (c) CMC-NT, and (d) CMC-NTC.

2.2. Force Field Development

In classical MD simulations, the force field is important because it defines the potential energy function that controls atomic motions and interactions. To compute the force field parameter, the Amber14SB force field[42] is used to describe both bonded and nonbonded interactions. It is an improved version of AMBER99SB,[43] which is widely applied for macromolecule simulations.[19−23,44−51] To compute the electrostatic component of nonbonded interactions, atomistic partial charges are assigned using the Gasteiger method.[52] These charges are necessary to compute the Coulombic interactions between atoms. The Amber14SB force field equation is given as:

$$V_{\text{total}} = \sum_{\text{bonds}} k_b(r - r_0)^2 + \sum_{\text{angles}} k_\theta(\theta - \theta_0)^2 + \sum_{\text{dihedral}} k_\chi[1 + \cos(n\chi - \delta_\chi)] + \sum_{\text{improper}} k_\psi[1 + \cos(n\psi - \delta_\psi)] + \sum_{i=1}^{N-1}\sum_{i<j}^{N}\left\{4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \frac{q_i q_j}{4\pi\varepsilon_0}\right\} \quad (1)$$

where V_{total} is the cumulative potential energy, encompassing contributions from bond elongation, angle deformation, torsional energy (both proper and improper dihedrals), van der Waals forces, and Coulombic interactions. The parameters include K_b (bond stretching constant), r (bond length), r_0 (standard bond length), k_θ (bending angle constant), θ (angle between molecules), θ_0 (standard angle), k_χ (proper torsional energy coefficient), χ (torsional interplanar angle), n (multiplicity), δ_χ (phase shift), k_ψ (improper torsional energy coefficient), ψ (improper torsional interplanar angle), δ_ψ (phase shift), ε_{ij} (depth of potential wall), σ_{ij} (distance where potential is zero), r_{ij}^{-12} (repulsive term dominant at short distance), r_{ij}^{-6} (attractive term dominant at long distance), ε_0 (permittivity of vacuum), q_i, q_j (charge within the two particles), and r (interspatial separations).

2.3. Immersion in the Aqueous Phase

The principal aim of this study is used to explore the dynamic behavior of CMC and its thiol-norborene-modified polymer derivative hydrogel models. To study the hydrogel model, four distinct polymer models are placed in a generously sized simulation box and immersed in water. To perform this operation, the Packmol open-source package is used.[53] The TIP4P water model is employed for the solvent molecules.[54] The present configuration of the system is delineated in Table S1, specifying its initial size and molecular composition.

2.4. MD Simulation

To perform MD simulations, the GROMACS 2022.3 package is used to generate the time-dependent trajectory through the MD simulation.[55] The system exhibits dynamic behavior by

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undergoing energy minimization, equilibration, and continuous production runs. Energy minimization employs the steepest descent algorithm, using an energy threshold of 10 kJ/mol· nm to ensure smooth transitions between MD simulation stages. Energy minimization is conducted with a step size of 0.001 over a maximum of half a million steps. Short-range van der Waals interactions are calculated with a 1.0 nm cutoff, whereas long-range Coulombic electrostatic interactions utilize a 1.2 nm cutoff. The long-range electrostatic interactions are being computed utilizing the particle mesh Ewald technique.[58] Periodic boundary conditions are employed across all three dimensions. The Leapfrog algorithm is used to integrate the equation of motion integration for the equilibration and production run with a step size of two fs. The equilibration process is currently being conducted within the canonical (NVT) ensemble for a duration of five ns, followed by an additional five ns within the isobaric isothermal ensemble (NPT). Throughout the equilibration and production runs, the system's temperature is maintained at a constant value of 300 K via the implementation of the velocity-rescale algorithm.[57] The LINCS algorithm[58] is implemented to restrict the vibrational degrees of freedom pertaining to hydrogen-containing groups throughout the equilibration procedure. During the NPT equilibration process, the pressure of the systems remains constant at 1.013 bar using the Parinello-Rahman algorithm.[59] The production run is performed in the NPT ensemble for 20 ns. The chronological trajectories are recorded at intervals of ten ps for the purpose of statistically averaging material properties. To visualize the chronological trajectory, the visual molecular dynamics (VMD)[60] package is used. The final snapshot of each system is given in Figure S3 of the Supporting Information. The chronological snapshot of the different systems is given in Figures S4−S7 of the Supporting Information. The systems are analyzed utilizing custom Python scripts.

2.4.1. Analysis of Trajectory. *2.4.1.1. Hydrogen Bonding.* Hbond interactions between polymer networks and solvent networks play a crucial role in determining the mechanical properties and swelling behavior of these gel networks. Multiple factors, like polymer nature, cross-linking degree, and functional moieties, influence Hbond strength. Hbond is a type of intermolecular interaction wherein a hydrogen atom is shared between two electronegative atoms, such as oxygen and nitrogen. In the case of these gel networks, the hydrogen bond occurs between the hydroxyl moieties of the polymer chains and water molecules present in the solvent. This interaction is vital in stabilizing the gel structure and controlling its properties. The Hbond forms between the atom acceptor and the atom donor based on a geometrical criterion. The formation of a Hbond occurs when the angle between the hydrogen donor and acceptor falls below 30°, and the donor and acceptor atoms exhibit an interatomic distance of under 0.35 nm. Using the sporadic Hbond autocorrelation function, Hbonding dynamics are evaluated as

$$C_{HB} = \frac{\langle H(\tau)H(\tau + t)\rangle}{\langle H(\tau)H(\tau)\rangle} \tag{2}$$

Where C_{HB} represents the Hbond count, and $H(t)$ takes values of 1 or 0 based on Hbond formation for the chosen pair at that instant.

2.4.1.2. Swelling Ratio. The swelling ratio is an important hydrogel characteristic that helps determine the solvent and drug release patterns in polymeric networks. The water absorption through the polymer network directly increases the hydrogel's weight, consequently raising the swelling ratio. The integration of freezing water and nonfreezing water is one of the two ways that polymers absorb water. Water molecules link to the chain of polymers by Hbonds are referred to as nonfreezing water. The interactions between water and the polar functional moieties of the polymer framework during freezing are relatively weak. The amount of water molecules that are directly linked to nonfreezing water by Hbonds serves as a proxy for its.[61] Both water types are employed to determine the swelling ratio, which can be computed as follows:

$$\mathrm{SR} = \left(\frac{W_{eq} - W_i}{W_i}\right) \times 100\% \tag{3}$$

After water swelling, the total weight of the hydrogel scaffold is indicated as W_{eq}, while W_i represents the weight of dried hydrogel composites.

2.4.1.3. Radial Distribution Function (RDF). The RDF is frequently utilized in MD simulations to characterize the particle spatial distribution. It provides valuable information about the system structure and interactions by assessing particle probability at defined distances from one another. The RDF is determined as

$$g(r) = \frac{1}{N}\left(\frac{V}{4\pi r^2\, \mathrm{d}r}\right)\sum_{i=1}^{N}\sum_{i\neq j}^{N}(|r_i - r_j| - r) \tag{4}$$

Where the RDF, symbolized as $g(r)$, quantifies the particle distribution, computed by considering the ratio of particles at distance r from a reference particle to the total N particles in the system within radial bins of width dr. Utilizing the Dirac delta function $\delta(z)$ to yield $z = 1$ or $z = 0$, distance $|r_i - r_j|$ represents the separation among the particles i and j. The computation of the RDF entails discretizing the simulation volume into successive concentric spherical layers with a fixed dr width. Within each layer, particle enumeration occurs, and subsequent normalization by the system's density yields $g(r)$. The RDF provides significant insights into the system configuration and interconnections. The coordination number establishes the proximate atoms or molecules encircling a central atom, exerting an effect on the immediate milieu and characteristics. It is determined as

$$\mathrm{CN}(r) = 4\pi\rho N\int_0^r g(r)r^2\, \mathrm{d}r \tag{5}$$

In this context, $\mathrm{CN}(r)$ signifies the coordination number, while ρ signifies the particle density, N corresponds to the total particle count, and $g(r)$ stands for the RDF. The integral over a spherical shell with a radius r measures particles located within that distance from the reference particle.

2.4.1.4. Mean-Squared Displacement. Self-diffusion assumes a pivotal role in the progression of microgel formation, ascertained through the derivation of the MSD slope over an extended temporal span. The MSD delineates the average particle migration from their respective initial locations, computed by squaring the positional variance at specified time intervals relative to their respective initial positions and subsequently averaging across all system particles. It is determined as

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$$\mathrm{MSD}(t) = \langle (r_i(t) - r_i(0))^2 \rangle = \frac{1}{N} \sum_{i=1}^{N} [(r_i(t) - r_i(0))^2] \tag{6}$$

The spatial coordinates of the i_{th} atom at time t are designated as $r_i(t)$, with the initial position indicated by $r_i(0)$. The collective mean over all particles within the system is denoted by angular brackets, ($\langle\ \rangle$). The logarithmic MSD slope (β) defines the diffusive characteristics. If $\beta = 1$ then it is characterized as a self-diffusive region. From the self-diffusive region utilizing the Einstein relationship, the diffusion coefficient D can be calculated from the slope of the MSD.

$$\beta = \frac{\log\left(\frac{\mathrm{MSD}(t)_2}{\mathrm{MSD}(t)_1}\right)}{\log\left(\frac{t_2}{t_1}\right)} \tag{7}$$

$$D = \frac{1}{6} \lim_{t\to\infty} \frac{\mathrm{d}(\mathrm{MSD}(t))}{\mathrm{d}t} = \frac{1}{6N} \lim_{t\to\infty} \frac{\mathrm{d}}{\mathrm{d}t} \sum_{i=1}^{N} [(r_i(t) - r_i(0))^2] \tag{8}$$

2.5. Data Rescaling

Data rescaling is applied to normalize the structural and transport properties within the range of $[-1$ to $1]$, ensuring that all features contribute on a comparable scale and eliminating distortions caused by differences in magnitude. This step is important because unscaled data can lead to biased analysis, reduced accuracy, and instability in predictive models. The mathematical transformation used is

$$x' = \left(\frac{x - x_{\min}}{x_{\max} - x_{\min}}\right)(x_{\max,\mathrm{new}} - x_{\min,\mathrm{new}}) + x_{\min,\mathrm{new}} \tag{9}$$

Where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the input data, respectively. $x_{\min,\mathrm{new}}$ and $x_{\max,\mathrm{new}}$ are the new minimum and maximum values, respectively.

2.6. Time-Dependent Structural and Transport Property Prediction

2.6.1. Echo State Network. The ESN is a form of RNN that belongs to the framework of reservoir computing. It consists of an input layer, a large recurrent reservoir of nonlinear neurons, and an output layer. The schematic representations of ESN are given in Figure 3. The input nodes are important for encoding external signals into the network; the reservoir nodes create a high-dimensional dynamic representation through recurrent activity, and the readout nodes transform this representation into the final output. The recurrent connections within the reservoir are randomly initialized. The reservoir provides dynamic mapping of the input into a high-dimensional space, which allows the network to capture temporal dependencies and nonlinear relationships in sequential data. The reservoir node connections are controlled by internal and external equations. The equations are given as

$$x[t+1] = (1 - \mathrm{lr})\cdot x[t] + \mathrm{lr}\cdot(W_{\mathrm{in}}\cdot u[t+1] + c_{\mathrm{in}}\cdot\xi) + W\cdot x[t] + W_{\mathrm{fb}}\cdot(g(y[t]) + c_{\mathrm{fb}}\cdot\xi) + b + c\cdot\xi \tag{10}$$

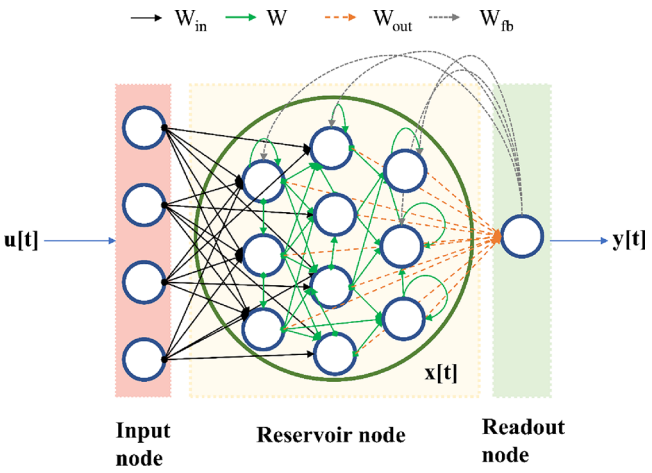


Figure 3. Schematic diagram of the ESN.

$$r[t+1] = (1 - \mathrm{lr})\cdot r[t] + \mathrm{lr}\cdot(W_{\mathrm{in}}\cdot u[t+1] + c_{\mathrm{in}}\cdot\xi) + W\cdot x[t] + W_{\mathrm{fb}}\cdot(g(y[t]) + c^{\mathrm{fb}}\cdot\xi) + b \tag{11}$$

$$x[t+1] = f(r[t+1]) + c\cdot\xi \tag{12}$$

Where $x[t]$ is the reservoir state vector at time t, which is crucial for maintaining temporal memory and dynamics, and lr is the leaking rate that controls how much of the previous state is retained versus the influence of new input, with its value varying between 0 and 1. The lr = 0 signifies that the reservoir state never updates and stays frozen. The lr = 1 signifies that the reservoir state fully updates at each step with no smoothing from the past. The values for lr range from 0 to 1, which is essential for temporal integration. The f and g are the reservoir activation functions, which help to understand the nonlinearity of the data. The activation functions considered in this work are tanh, sigmoid, softplus, and identity. The W_{in} is the input weight matrix that projects external inputs into reservoir space, which determines the input sensitivity, $u[t+1]$ is the input signal at the next time step, c_{in} is the input noise scaling factor, $\boldsymbol{\xi}$ represents random noise for regularization and robustness, and W is the recurrent weight matrix defining internal reservoir connections (the heart of reservoir dynamics and memory). The recurrent weight matrix is randomly initialized by the Python code. The reservoir connectivity ranges between 0.1 and 1, where 0.1 signifies that 10% of the connections are nonzero. The W_{fb} is the feedback weight matrix that incorporates output feedback for closed-loop operation (important for autonomous behavior), $y[t]$ is the feedback vector from a previous time step, c_{fb} is the feedback noise scaling, b is the bias vector providing baseline activation, and c is the reservoir noise scaling factor. The noise parameters are typically small positive floats. The noise range of 0 to 5 has been utilized for this study. The bias matrix is randomly initialized. The r represents the internal activation vectors, which are used for the external equations. These equations are given in eqs 10−12.

The third node of the reservoir is the readout node. The readout node is the sole trainable component in the ESN framework, responsible for mapping the reservoir's high-dimensional state to the desired output.[62−64] It is often used with linear methods to train the readout node. This design reduces the computational cost and trains faster compared to the traditional RNN framework. The ESN framework avoids the need for back propagation through time, which reduces training

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time. It uses a simple linear regression on the output layer, which significantly lowers the computational cost compared to conventional recurrent networks. Using the linear regression methods extract spatiotemporal patterns from the reservoir's dynamics, which act as a nonlinear excitable medium. This design allows the same reservoir to be reused for diverse tasks by retraining only the readout.[62] Ridge regression (Tikhonov regularization) is the preferred method for training ESN readouts due to its robustness. It modifies standard linear regression by adding an L2 regularisation term. For this study, the Ridge regression values are set to range between 0 and 1. A regularization value of 0 in the Ridge regression signifies that no regularization is being applied. This makes the model equivalent to a standard Ordinary Least Squares (OLS) linear regression model. In this case, the model's objective is solely to minimize the sum of squared errors, without any penalty on the size of the coefficients. This can lead to overfitting. A regularization value of 1 represents a specific, nonzero amount of penalty applied to the model's coefficients. It is a value that shrinks the coefficients toward zero, reducing the model's complexity and its tendency to overfit. The mathematical formulas for the readout layer are given as

$$W_{\text{out}} = YX^{\text{T}}(XX^{\text{T}} + \lambda Id)^{-1} \tag{13}$$

$$y = W_{\text{out}}^{\text{T}}x + b \tag{14}$$

Where X denotes the matrix of input state vectors collected during training, Y represents the corresponding target outputs, λ is the ridge parameter controlling regularization, and I is the identity matrix ensuring numerical stability. After obtaining the W_{out}, the model prediction is expressed in terms of eq 14. The x represents the new input vector, b represents the bias term, and y represents the predicted output vector.

Hbond and MSD data are extracted from the MD trajectory and used as inputs to the ESN framework. Due to differences in their patterns, two separate ESN models are trained. In addition, the data scales vary among different materials. To ensure consistency within a defined range, all data are normalized to values between −1 and 1. The normalized data feeds into the ESN framework, where the flow of information is organized in a structured way to generate predictions. At the beginning, the incoming data first pass through the input node, where each signal is multiplied by an input weight matrix that adjusts the scale of influence before moving further into the system. The processed data then enter the reservoir, which is a dynamic structure containing recurrent connections. These recurrent loops allow the current signal to combine with previous internal states, which means that the system is able to remember and capture patterns from both recent and past data. This property helps in creating a rich and higher-dimensional representation of the input sequence without explicitly training the inner connections. The reservoir activity evolves naturally as new information flows, generating diverse internal responses. Once this dynamic state representation is formed, it is transferred to the readout structure. The readout structure is the only part that undergoes training and learns to map the evolving reservoir states to the required output.

2.6.2. Gated Recurrent Unit (GRU). The GRU is a variant of the RNN that has been designed to model sequential data in an efficient way and capture long-term dependencies. Compared to the traditional RNN, which has the vanishing or exploding gradients problem when it deals with long sequences, the GRU introduces the gating mechanism to control the flow of information. While the ESN uses a fixed reservoir, the GRU uses a procedure to optimize all the weights on the inside in the training procedure as back propagation through time (BPTT). The GRU simplifies the architecture of the LSTM network by combining the two gates (forget and input) into a single one (named "update") and combining the cell state with the hidden state. There are two main gates in the architecture, the update gate and the reset gate. The update gate is responsible for controlling how much information from the past should be passed to the future, while the reset gate controls how much information from the past must be discarded. The governing equations for the GRU cell are well recorded in the literature.[65] The candidate hidden state is calculated in this framework using the MD property and reset gate, so that the model will ignore the irrelevant past information. The last hidden state is a linear interpolation between the preceding state and the candidate state adjusted by the update gate. This mechanism allows the GRU to retain the memory of relevant features over time without the computational cost of a full LSTM.

The rescaled Hbonding and MSD data are fed into the GRU framework for making forecasts for the future. The model is trained by minimizing the root-mean-square error (RMSE) between predicted and actual MD-generated values as follows. The hyperparameters, such as number of layers, hidden size, learning rate, batch size, optimizer, epoch, and percentage of data, are optimized to get the highest coefficient of determination (R^2), and the lowest RMSE.

2.6.3. Long Short-Term Memory. The LSTM network is the specific RNN architecture that has been designed in order to solve the problems of normal RNNs, i.e., vanishing and exploding gradient problems when learning long-term dependencies. Unlike the normal RNN, where the gradients can destabilize over long sequences of data, the LSTM allows for a complex memory cell controlled by gating mechanisms. These gates control the flow of information, enable the network to maintain relevant information for long durations of time, and expel extraneous information. The essential feature of the LSTM is a memory cell state, which can be viewed as a conveyor belt that can pass information across time steps with minimal interaction (linear function). Information flows into and out of the cell state though three different gates: the forget gate, the input gate, and the output gate. These gates use the sigmoid activation functions to produce values ranging between 0 and 1, specifying the amount of information propagated. The mathematical formulations of the LSTM unit are well documented in the literature.[66]

For this study, the LSTM framework is trained on the rescaled Hbonding and MSD data taken from the MD trajectory. Training is done through BPTT to obtain the smallest possible value for the RMSE between the predicted and the actual values. Key hyperparameters are optimized such as the number of layers, hidden size, learning rate, batch size, optimizer, epoch, and data percentage in order to ensure that the model accurately captures the nonlinear temporal dynamics of the hydrogel system.

2.6.4. Lag Large Language Model Meta AI (LAG-LLAMA). The LAG-LLAMA framework is a paradigm shift from data set-specific recurrent models to a general-purpose foundation model architecture for univariate probabilistic time series forecasting. While ESN, GRU, and LSTM models process time series data sequentially using hidden states that carry information from previous steps. In contrast, LAG-LLAMA uses a decoder-only transformer that captures long-range depend-

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encies directly through attention mechanisms. One of the most important innovations in the LAG-LLAMA framework is tokenization. Unlike conventional transformers that can take time series points as discrete tokens or patches, LAG-LLAMA builds "lag features" as covariates. For a given time step, the input token is created by retrieving values from certain prior time steps (lags) following a set of indices (e.g., quarterly, monthly, daily, hourly offsets). The definition of a time series lag operation is the mapping onto a vector, where each entry is defined in the literature.[67] These lag features are concatenated with date/time covariates to create the input token, which lets the model implicitly learn about the frequency and periodicity of data. The architecture includes the pre-normalization step by using root-mean-squared layer normalization over a mini-batch of inputs (RMSNorm) and the use of rotary positional encoding (RoPE) in every attention layer to manage the sequentiality of the data and the invariance properties of time series. Instead of delivering a single deterministic quantity, LAG-LLAMA is aimed at probabilistic forecasting. The final layer in the network is a "distribution head," which projects the transformer features into the parameters of a probability distribution.

The rescaled Hbond and MSD data are fed into the model to predict the future data. The hyperparameters for LAG-LLAMA, including the number of layers, model dimension (d_model), number of attention heads (n_heads), learning rate, batch size, optimizer, epoch, and data percentage are optimized to minimize the RMSE for the specific Hbond and MSD data sets.

2.6.5. Analysis of the Echo Static Neural Network Result. *2.6.5.1. Root-Mean-Squared Error (RMSE).* RMSE measures the average magnitude of the difference between predicted values and actual values. It gives an idea of how well a model fits the data, with lower values indicating better performance.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_{\mathrm{actual}} - y_{\mathrm{predicted}})^2} \tag{15}$$

Where y_{actual} is the actual value, which comes from the MD trajectory, $y_{\mathrm{predicted}}$ is the predicted value from the ESN framework, and n is the total number of data points

2.6.5.2. Coefficient of Determination (R^2). The R^2 shows how much of the variance in the dependent variable can be explained by the model. Its typical value ranges between 0 and 1, where values closer to 1 indicate a better fit. Occasionally, R^2 takes on a negative value, indicating that the model is performing worse than a simple baseline model that consistently predicts the mean of the actual data:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_{\mathrm{actual}} - y_{\mathrm{predicted}})^2}{\sum_{i=1}^{n}(y_{\mathrm{actual}} - \overline{y}_{\mathrm{actual}})^2} \tag{16}$$

Where y_{actual} is the actual value, which comes from the MD trajectory, $y_{\mathrm{predicted}}$ is the predicted value from the ESN framework, and $\overline{y}_{\mathrm{actual}}$ is the mean of actual value.

3. RESULT AND DISCUSSION

3.1. Hyperparameter Optimization

3.1.1. Hyperparameter Optimization for Hbond Data. The AI frameworks such as ESN, GRU, LSTM, and LAG-LLAMA frameworks are used, which have several hyperparameters. In this work, a single hyperparameter is optimized at a time, while the other hyperparameters remain at their default values as specified by the code. The grid search method is used to find the optimum hyperparameter in the above architecture. After the first hyperparameter optimization, other hyperparameters that require optimization are subsequently addressed.

3.1.1.1. Optimization of ESN Framework in Hbond Data. Figure S8 and Tables S2−S12 show the optimization of the hyperparameters for Hbond data in the ESN network. Initially, a number of neurons are optimized for the ESN framework in the Hbond data. The R^2 value is negative for the first five neurons and then becomes positive up to around 100 neurons. Beyond this point, the R^2 value fluctuates around a mean and generally remains negative for up to 500 neurons. Within the first 100 neurons, the overall slope of R^2 is negative, although a small portion of the data shows a local linear increase. The highest R^2 value is observed in 38 neurons, reaching 0.90. When too many neurons are used, the ESN tends to memorize noise instead of learning the true patterns, which reduces its ability to generalize to new data. On the other hand, with too few neurons, the network does not have enough capacity to capture the dynamics, leading to poor performance and negative R^2 values. After optimizing the number of neurons, the lr hyperparameter is optimized. Initially, increasing the lr value, the R^2 is increasing until an lr value of 0.53, because a moderate leaking rate allows the reservoir to balance short-term memory and adaptability. After that, the lr value increases, the R^2 values decrease because too high a leaking rate makes the reservoir update too quickly, losing useful temporal information. After optimizing the lr value, the sr hyperparameter is optimized. Initially, sr = 0, the R^2 shows 0.99 because with no recurrent influence, the model behaves almost like a feedforward network, which can still fit simple temporal trends. However, after increasing the sr value from 0 to 0.1, the R^2 decreases because small recurrent feedback introduces instability without adding enough memory capacity. After 0.1, the increasing *sr* value, the R^2 starts to increase until a value of 0.96, where the R^2 value shows maximum because a suitable spectral radius enhances the reservoir's memory and nonlinear dynamics. After that, as the *sr* value increases, the R^2 decreases in nature because a too-large spectral radius makes the reservoir chaotic, reducing prediction accuracy. After optimization of the *sr* hyperparameter, the input scaling hyperparameter is optimized. Initially, its value is set to 1. As the hyperparameter decreases, the R^2 value starts to increase because smaller input scaling prevents the reservoir from being saturated and allows the dynamics to capture finer variations in the input signal. The R^2 value is observed at 0.66; after that, decreasing input scaling, the R^2 also decreases because too small scaling weakens the influence of the input signal, reducing the reservoir's ability to learn meaningful dynamics.

After optimizing the input scaling hyperparameter, the noise in the input node, the noise in the reservoir node, and the noise in the feedback dimension are optimized, which are shown in Tables S2−S4 of Supporting Information. Initially, the noise is set to 0 by default, giving an R^2 of 0.99. As the noise level increases, the R^2 value starts to decrease. This reduction occurs because noise introduces random variations into the reservoir states, which interfere with the model's ability to capture the true underlying dynamics and lead to less accurate predictions. However, when noise is added specifically to the feedback dimension, the performance of the ESN framework remains unaffected. This is because the feedback signal is already regulated by the reservoir dynamics, and small perturbations in this dimension are smoothed out, preventing them from

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significantly influencing the overall learning process. After optimization of noise hyperparameters, the input bias and feedback scaling hyperparameters are optimized, as shown in Tables S5−S6 of Supporting Information. Initially, the input bias is set to TRUE by default, giving an R^2 of 0.99. As it is set to FALSE, it is also showing, but the R^2 value remains the same, but the RMSE decreases. This happens because removing the bias term reduces unnecessary offsets in the input signals, allowing the ESN network to represent the H-bond data more efficiently and improving error minimization without affecting the overall fit. After that, the feedback scaling hyperparameter. This hyperparameter does not affect the performance of the ESN framework for Hbond data. After that, the input node connectivity, reservoir node connectivity, and feedback node connectivity of hyperparameters are optimized, which are shown in Tables S7−S9 of Supporting Information. Initially, these connectivity values are set to 0.10 by default, giving an R^2 value of 0.99. As the input connectivity increases, the R^2 value remains close to 0.99, but the RMSE decreases until reaching a connectivity of 0.36. This improvement occurs because moderate input connectivity ensures that a sufficient number of input signals are transferred to the reservoir, enriching the dynamics without adding redundancy. When the input connectivity increases beyond this point, the *RMSE* starts to rise as too many input links inject excess information into the reservoir, creating overlapping or noisy state representations. For reservoir connectivity, as it increases, the R^2 begins to decrease since higher internal connectivity reduces reservoir sparsity and leads to stronger correlations among nodes, which limit the reservoir's ability to generate diverse and independent dynamics. Finally, feedback connectivity does not affect the performance of the ESN network because the reservoir already integrates feedback signals effectively, and additional connectivity in this dimension does not contribute new information to the learning process.

Afterward, the activation functions of the reservoir node and the reservoir equation are optimized, as shown in Tables S10−S11 of the Supporting Information. For the Hbond dynamics, the tanh, sigmoid, softplus, and identity activation functions are tested. Among them, the lowest RMSE is observed with the Tanh activation function. This is because tanh is a smooth, nonlinear, and symmetric function that maps inputs to the range of −1 to 1, which helps the reservoir capture both positive and negative fluctuations in the Hbond dynamics. As the equation hyperparameters are optimized in the ESN framework, the R^2 value remains around 0.99 for both the internal and external equations, which are given in eqs 9−11. The RMSE is lower for the internal equation of the ESN framework because the internal update directly governs the reservoir state dynamics, allowing it to represent the temporal dependencies of the input data more effectively. Afterward, the readout node hyperparameter is optimized. The ridge values are varied from 1×10^{-1} to 1×10^{-15}. Among them, the lowest RMSE is observed at 1×10^{-6}, because this value provides an optimal balance between bias and variance large enough to regularize the model and prevent overfitting but small enough to preserve the essential patterns in the Hbond data. After that, the RMSE values start to increase, as very small ridge values (closer to zero) weaken the regularization effect, causing the model to overfit noise in the reservoir states and lose generalization ability. The final hyperparameters of the ESN framework are shown in Table 1.

3.1.1.2. Optimization of the GRU Framework, Hbond Data. Figure S9 and Tables S13−S18 show hyperparameter optimization of Hbond data in the context of GRU network. At the beginning, the number of layers is optimized for the GRU framework. The R^2 value shows that it has a low positive value of 0.02 for the first three layers. Beyond this point, the value of the R^2 is brought to a negative signal as the number of layers increases up to 10, which shows that a deeper network structure causes overfitting or vanishing gradient problems, thus hindering the temporal learning of Hbond data. The lowest RMSE of 0.28 is seen for 3 layers. After optimization on the number of layers, the hyperparameter hidden size is optimized. The RMSE decreases with the increase in the size of the hidden unit and reaches the minimum (0.28) with the size of the hidden unit = 128. Beyond this point, increased hidden size causes performance to decrease, as a too-large hidden state introduces redundancy and noise into the model. Subsequent optimization of the learning rate, it is observed that R^2 value is low but hits a high value, i.e., 0.03 when the learning rate is set to 7.00×10^{-3}. Lower learning rates achieve slow convergence, and significantly higher learning rates, such as 1.00×10^{-02}, cause the model to overshoot the optimum, and achieve a large negative calculation of R^2, which is −1.03. The batch size is then optimized, and the lowest RMSE (0.27) is achieved at 128. Smaller batch sizes add instability to the gradient estimation, and this particular batch size makes the update for the GRU weights stable. Different optimizers are subsequently tested, AdamW optimizer had the highest R^2 of 0.04 and a low RMSE of 0.27. This implies that the

Table 1. Optimized ESN Parameters for Hbond and MSD Data

parameter[a]	Hbond	MSD
units	38	6
lr	0.53	0.20
sr	0.96	0.60
input_bias	false	true
input_scaling	0.66	0.16
noise_in	0	0
noise_fb	0	0
noise_rc	0	0
noise_type	normal	normal
bias_scaling	1.0	1.0
fb_scaling	1	1
input_connectivity	0.36	0.10
rc_connectivity	0.10	0.10
fb_connectivity	0.10	0.10
equation	internal	internal
activation	tanh	tanh
ridge	1×10^{-6}	1×10^{-7}
percent of data	10	10
RMSE	10.58×10^{-3}	1.58×10^{-3}
R^2	0.99	0.99

[a]units = number of the neurons = range of 1−500; lr = leaking rate = range of 0−1; sr = spectral radius = range of 0−5; input bias = true or false; input scaling = range of 0−5; noise_in = input noise = range of 0−5; noise_fb = noise in feedback dimension = range of 0−5; noise_rc = noise in reservoir = range of 0−5; noise type = normal or uniform; bias_scaling = range of 0−5; fb_scaling = feedback scaling = range of 0−5; input_connectivity = input node connectivity = range of 0−1; rc_connectivity = reservoir connectivity = range of 0−1; fb_connectivity = feedback dimension connectivity = range of 0−1; equation = reservoir node computing formulas = internal or external; activation = activation function in reservoir node = tanh or sigmoid or softplus or identity; ridge = the readout node = range of 1×10^{-1} to 1×10^{-15}.

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decoupled weight decay in AdamW better regularizes the model than Adam, RMSprop or SGD for this data set. Last, the percentage of training data is optimized. The model produced its best performance with 60% of the data, R^2 = 0.05 and RMSE = 0.24.

3.1.1.3. Optimisation of LSTM Framework Using Hbond Data. Figure S10 and Tables S19−S24 show the optimization of hyperparameters for Hbond data inside of the LSTM network. Initially, the number of layers is optimized for the LSTM framework. The value of R^2 shows a low positive value of 0.02 in the first two layers. However, when the number of layers is increased from 3 to 10, the R^2 value turns negative. Thus, the network becomes more complex, and it is getting overfitted, which prevents the model from learning the correct Hbond patterns in the network. With two layers, the lowest RMSE of 0.29 is obtained. After optimizing the number of layers, the hyperparameter hidden size is optimized. The RMSE drops slightly with an increase in the hidden size, and the lowest value of the RMSE (0.28) is associated with hidden sizes of 512. At a size of 1024, there seems to be a drop in performance as R^2 becomes negative. This negative R^2 suggests that too many hidden units introduce noise. Subsequent optimization of the learning rate reveals that the lowest RMSE of 0.27 is achieved at a learning rate of 1.00×10^{-4}. Higher learning rates, e.g., 1.00×10^{-2} , result in major failure of the model, resulting in a large negative value of R^2, i.e., −0.67, where the model overshoots the optimal values during training. The batch size is then optimized, and the lowest RMSE value of 0.27 is achieved with a batch size of 8. Larger batch sizes, such as 256 and 512, give negative values for R^2, meaning that larger batches do not update the weights very well for the particular data set. Different optimizers are then tried, and the RMSprop optimizer is shown to be the most effective, with an RMSE of 0.26 and an R^2 of 0.03. This implies that the adaptive learning rate of RMSprop is better able to handle the Hbond data than Adam, AdamW, or SGD. Finally, the training data percentage is optimized. The model showed its strongest performance using 60% of the data with an R^2 value of 0.04 and an RMSE value of 0.25. Similar to the case of the GRU framework, the LSTM network has much lower predictive accuracy than the ESN framework for Hbond data.

3.1.1.4. Optimization of LAG-LLMA Framework in Hbond Data. Figure S11 and Tables S25−S32 show the optimization of the hyperparameters for Hbond data in the LAG-LLMA network. Initially, the number of layers is optimized. The R2 value remains negative across most layer counts, ranging from −0.02 to −0.15. This suggests that increasing the depth of the Llama-based architecture does not effectively capture the Hbond dynamics and introduces unnecessary complexity. The lowest RMSE of 0.29 is observed with 12 layers. After optimizing the number of layers, the d_model is optimized. The model achieves the best performance at a lower dimension of 8, with an RMSE of 0.28. Increasing the d_model to 128 or higher significantly increases the error and results in large negative R^2 values, indicating that high-dimensional representations introduce noise into the prediction. After optimizing the d_model, the number of attention heads (n_heads) is optimized. As the n_heads increase, the RMSE decreases until the value of 8. After that increasing the RMSE increases. Increasing n_heads up to 8 enhances model capacity by allowing parallel subspaces to capture diverse temporal dependencies in lag features, improving representation of complex patterns, and reducing prediction error (RMSE). Each head processes a portion of the d_model, enabling finer-grained attention over time series covariate features and summary statistics. Beyond 8 heads, RMSE rises due to smaller d_model, which reduces each head's representational capacity and increases optimization difficulty, leading to underfitting or poorer generalization on noisy time-dependent data. Next, the dropout rate is optimized. The lowest RMSE of 0.27 is observed at a dropout rate of 0.1. Following this, the learning rate is optimized. The best performance is found at a learning rate of 5.00×10^{-3}, with an RMSE of 0.26 and a slightly positive R^2 of 0.01. Extremely low or high learning rates lead to negative R^2 values, showing the model's sensitivity to gradient updates. Subsequently, the batch size is optimized. Smaller batch sizes of 4 and 32 yield the lowest RMSE of 0.26. The optimizer is then selected, with Adam showing the best stability (RMSE 0.26) compared to AdamW, RMSprop, or SGD. Finally, the data percentage for training is optimized. The model performs best with 70% of the data, achieving an R^2 of 0.02 and an RMSE of 0.26.

3.1.1.5. Comparison of ESN, GRU, LSTM, and LAG-LLMA Architectures for Hbond Data. This study compares the performance of the ESN, GRU, LSTM, and LAG-LLMA frameworks for predicting Hbond dynamics. The ESN framework demonstrates superior performance compared with the other three models. The ESN model achieves an R^2 value of 0.99 and a low RMSE of 0.01 using only 10% of the data for training. This indicates that the reservoir computing approach effectively captures the complex, nonlinear temporal patterns of H-bonding without extensive training. In contrast, the GRU framework shows a significantly lower accuracy. Even after optimizing layers, hidden size, and learning rates, the GRU model achieves a maximum R^2 of only 0.05 and the RMSE of 0.24. Similarly, the LSTM network performs poorly, with an optimized R^2 of 0.04 and an RMSE of 0.25. Both GRU and LSTM models struggle to learn the Hbond fluctuations. This suggests that the gating mechanisms in these deep learning models require larger data sets to converge and are prone to overfitting or vanishing gradient problems when applied to this specific short-trajectory MD data. The transformer-based LAG-LLAMA model also fails to outperform the ESN. The final optimized model stabilizes at a low R^2 of 0.02 and the RMSE of 0.26. The high dimensionality and complexity of the LAG-LLAMA architecture introduce noise and instability for this type of data set. Therefore, ESN proves to be the most efficient and accurate tool for predicting the future properties of hydrogels in this study.

3.1.1.6. Optimization of the ESN Framework in MSD Data. Figure S12 and Tables S33−S45 of the Supporting Information show the hyperparameter optimization of the ESN framework for the MSD data. Initially, the number of neurons is optimized for the ESN framework of MSD data, which is shown in Figure S12a of Supporting Information. For the first five neurons, the R^2 value is negative because the reservoir is too small to capture the linear trend of the MSD data, leading to poor fitting. At the sixth neuron, the highest R^2 value observed is 0.98, as the reservoir reaches the minimum size required to represent the linear increase in the MSD with time. As the number of neurons increases beyond this point up to around 38, the R^2 value decreases because the additional neurons introduce redundancy and noise, which weakens the model's ability to follow the simple linear behavior of MSD. When the reservoir size continues to increase beyond 38 neurons and up to 500 neurons, the R^2 value becomes negative again since the overparameterized network loses generalization and overfits random fluctuations instead of reproducing the linear MSD trend. Afterward, the lr hyperparameter is optimized, which is

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shown in Figure S12b of Supporting Information. As the lr value increases from 0.1 to 0.2, the RMSE value decreases because the reservoir updates more effectively, allowing the ESN to capture the smooth linear growth of the MSD data. At the lr = 0.2, the R^2 value reaches 0.99. After that, further increasing the lr value causes the RMSE to rise since the reservoir dynamics become too fast and fail to preserve the long-term correlations that are essential for reproducing the MSD trend. Afterward, the sr hyperparameter is optimized, which is shown in Figure S12c of Supporting Information. When sr = 0, the R^2 value is 0.99 because the reservoir behaves like a direct linear mapping, which matches the linear character of the MSD data. However, as the sr value increases from 0 to 0.1, the R^2 value initially becomes negative since the reservoir dynamics introduce instability and nonlinear behavior that does not align with the simple linear MSD trend. As the sr value increases to 0.4, the R^2 value becomes positive again because the reservoir dynamics start stabilizing and partially capture the temporal correlations in the MSD data. The lowest RMSE is observed at sr = 0.6 with a value of 20.88×10^{-3}, as this balance between reservoir stability and memory enables the network to best reproduce the linear MSD curve. After this point, further increasing the sr causes the RMSE to rise because excessive sr leads to overamplification of internal states, which adds noise and reduces accuracy in the ESN framework for the MSD data.

Afterward, the input scaling hyperparameter is optimized in the ESN framework for the MSD data, which is shown in Figure S12d of Supporting Information. Initially, the input scaling value is set to 1.0 by default, where the R^2 value reaches 0.99, since the network directly follows the linear MSD trend. When the input scaling is reduced to 0.1, the RMSE value decreases because the reservoir receives smoother and less amplified inputs, which improves stability in representing the linear growth of MSD. The lowest *RMSE* value is observed at an input scaling of 0.16 with a value of 1.48×10^{-3}, as this provides an optimal balance between input sensitivity and stability, allowing the ESN to capture the MSD behavior with high accuracy. However, when the input scaling value is further increased, the RMSE starts to rise because overly strong input signals drive the reservoir into unstable dynamics. Afterward, the noise of the input node, reservoir node, and feedback data hyperparameters is optimized, which is shown in Figure S12e and Tables S33 and S34 of Supporting Information. Initially, when the noise value is set to 0, the R^2 value is 0.99, because the ESN reproduces the linear MSD data without distortion. However, when noise is added to the input and reservoir nodes, the R^2 value decreases since random perturbations interfere with the reservoir's ability to capture the smooth linear trend of MSD. In contrast, adding noise to the feedback data does not affect the performance of the ESN framework because the MSD data is linear in nature, so the reservoir can still maintain accurate predictions even when minor noise is introduced through the feedback channel. Afterward, the input bias, bias scalln, and feedback scaling hyperparameters are optimized, which are shown in Tables S35−S38 of Supporting Information. Changing the hyperparameter values does not affect the training of the ESN framework for the MSD data. Afterward, the input node connectivity, reservoir node connectivity, and feedback connectivity hyperparameters are optimized, which are shown in Tables S39−S42 of Supporting Information. Initially, these hyperparameter default values are set to 0.1 with an R^2 of 0.99, as this sparse connectivity allows the ESN to reproduce the linear MSD data efficiently. However, increasing the connectivity value reduces the R^2 value, except for the feedback connectivity, because denser input and reservoir connections introduce redundant and noisy interactions that distort the simple linear trend of MSD. In contrast, increasing the feedback connectivity value does not affect the performance of the ESN framework since the MSD data evolve in a linear manner, and the model's ability to follow this trend does not depend on stronger feedback coupling.

Afterward, the activation functions of the reservoir and the equation reservoir are optimized, which are shown in Tables S43−S44. For the MSD data, the tanh, sigmoid, softplus, and identity activation functions are tested. Among them, the lowest RMSE is observed with the tanh activation function because it provides a symmetric nonlinearity that maps inputs to the range $[-1,1]$, which helps maintain stability in the reservoir states while preserving sensitivity to both positive and negative fluctuations in the MSD data. In contrast, the sigmoid and softplus functions are asymmetric and compress inputs into a strictly positive range, which reduces the reservoir's ability to represent the balanced linear growth of MSD. The identity function lacks nonlinearity, which limits the expressive power of the reservoir. Thus, tanh achieves the best balance between nonlinearity, stability, and dynamic range, making it most effective for reproducing the MSD trend. Afterward, the equation hyperparameters are optimized. Changing the equation hyperparameter does not affect the performance of the ESN framework. Afterward, the ridge hyperparameter is optimized, which is shown in Table S45 of Supporting Information. The range of the ridge parameter is studied from 1×10^{-11} to 1×10^{-15}. Among them, the lowest RMSE is observed at 1×10^{-7} with a value of 1.58×10^{-3} because this level of regularization provides the right balance between controlling overfitting and maintaining model flexibility for the linear MSD data. However, when the ridge value is increased further, the RMSE starts to rise since stronger regularization over smooths the output weights, reducing the ESN's capacity to capture the MSD trend accurately.

3.1.1.7. Optimization of GRU Network in MSD Data. The systematic tuning for the hyperparameters of MSD data in the GRU network is documented in Figure S13 and Tables S46−S51. Initially, the number of layers is optimized. The model is unstable due to the number of layers used to fit the model. On the opposite extreme, having three layers results in a positive R^2 of 0.57 and the lowest value of 0.11 for RMSE. Augmenting the number of layers past this point, from four to ten layers, leads to negative R^2 and high RMSE values, suggesting that a deep network cannot capture the linear trend of the MSD data. After the optimum number of layers is determined, the hidden size is optimized. The model does not do well at low values of hidden sizes, and from sizes 2 to 32, the results are poor. A very substantial improvement occurs at a hidden size of 64 units, where the R^2 reaches 0.92. The minimum RMSE value of 0.03 is obtained for hidden size 1024, suggesting that larger is better for this particular data set. Next, the learning rate is optimized. Extremely low learning rates (1.00×10^{-6} to 1.00×10^{-5}) result in large negative R^2 values. The maximum performance is given by 8.00×10^{-4} learning rate, with a high R^2 = 0.95 and a low RMSE = 0.04. Afterward, the batch size is optimized. Small batch sizes of 4 and 8 produce negative R^2. The most stable and accurate configuration is at a batch of 128 units, resulting in an RMSE of 0.02 and R^2 is 0.95. Following this, different optimizers are studied. The Adam optimizer turns out to be the best and reaches the R^2 value of 0.95 and the RMSE value of 0.02. In

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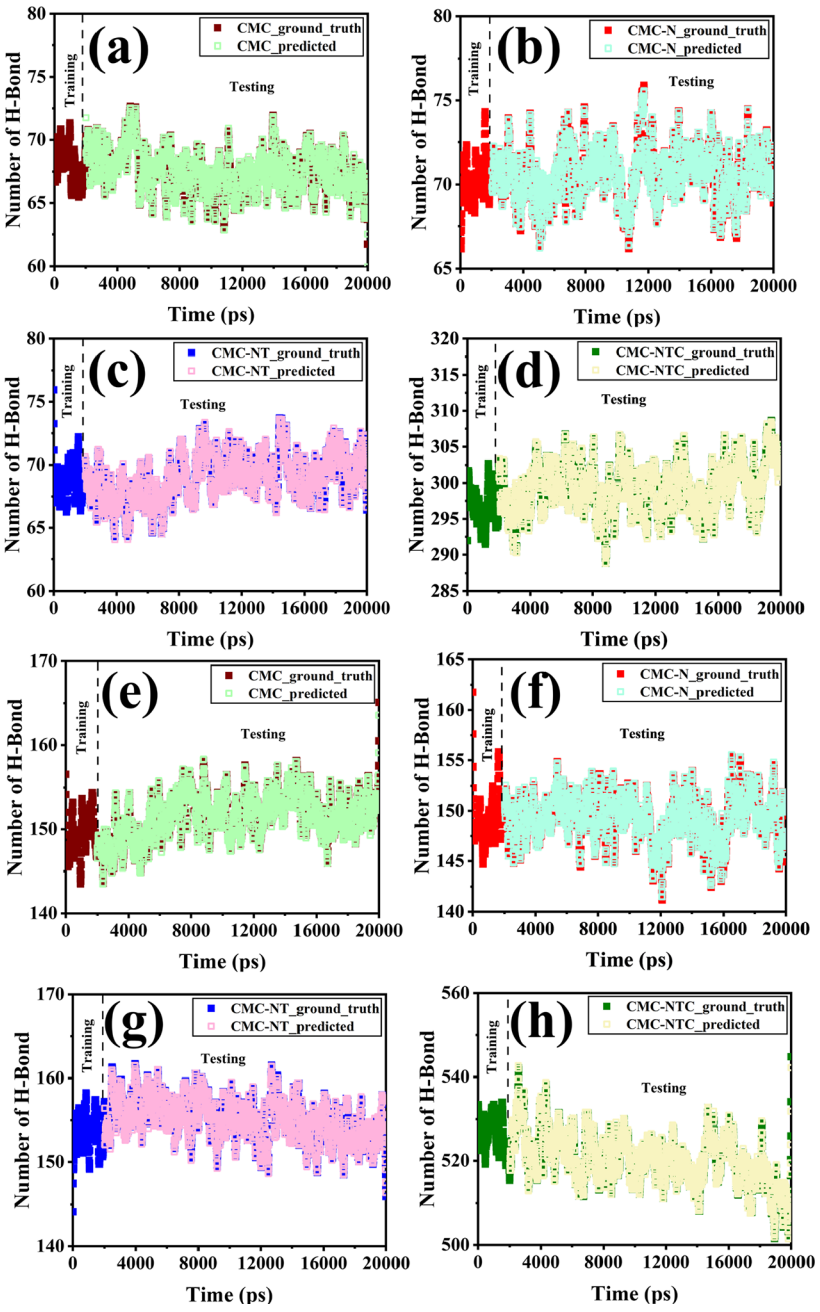


Figure 4. Hbond count within the system (a) CMC to CMC, (b) CMC-N to CMC-N (c) CMC-NT to CMC-NT, (d) CMC-NTC to CMC-NTC, (e) CMC to solvent, (f) CMC-N to solvent (g) CMC-NT to solvent, and (h) CMC-NTC to solvent.

comparison, AdamW, RMSprop, and SGD do not model the data properly, and this is verified by negative values of R^2. Finally, the proportion of data to be fed for training is optimized. The best performance of this model is observed for 70% of the data with an R^2 value equal to 0.95 and RMSE equal to 0.02. This suggests that, contrary to the case with Hbond data for which the GRU framework categorically breaks down, the GRU architecture can successfully predict MSD data if hyperparameters for the framework are meticulously optimized.

3.1.1.8. Optimization of LSTM Framework in MSD Data. Figure S14 and Tables S52 and S57 show the systematic tuning of hyperparameters for MSD data in the LSTM network. The number of layers is optimized in the beginning. The model shows very acceptable performance using one layer with an RMSE of 0.02 and R^2 0.93. However, adding too much depth to the network leads to catastrophic failure; with three-ten layers, incredibly negative R^2 values of −162.67 to −428.54 are obtained. This indicates that the deep LSTM architecture adds too much complexity that affects its capacity to learn the comparatively simple, linear progression of the MSD data causing severe overfitting. After the optimum number of layers is determined, the size of hidden layer is optimized. Small values of hidden size (2−32) result in negative R^2 values, which means the network is not able to learn the trend. Performance is greatly increased at hidden sizes of 64 (R^2 = 0.94) and 128 (R^2 = 0.93). However, increasing the hidden size further to 1024 decreases the performance (R^2 = −1.61) as the network becomes disproportionate and is looking at the noise, not the actual signal. Next, the learning rate is optimized. Extremely low learning rates (1.00×10^{-6}) produce large negative R^2 values.

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That is, the best possible range is between 1.00×10^{-3} and 4.00×10^{-3}, in which the model will achieve R^2 values of up to 0.95. Higher learning rates of values such as 1.00×10^{-1} reduce performance. Thereafter, the batch size is optimized. The model works better for smaller batch sizes; the lowest RMSE and R^2 values equal to 0.01 and 0.97 are seen for batch size 16. Increasing the batch size to 128 gives a negative R^2 value of −12.60, which implies that the rarity of weight update hinders convergence for this data set. Following this, different optimizers are checked. The Adam and AdamW optimizers provide the best results with an R^2 of 0.99 and RMSE of 0.01 to 0.02. On the other hand, SGD is not very good, achieving an R^2 of −47.01. Finally, the percentage of the data to train is optimized. The LSTM framework is found to be very efficient for the MSD data as they are able to reach the R^2 value of 0.99 with only 40% of the training data. This shows that, unlike what is possible for Hbond data, the LSTM network knows how to predict MSD dynamics if the hyperparameters are well-tuned.

3.1.1.9. Optimization of LAG-LLMA Framework in MSD Data. Figure S15 and Tables S58−S65 optimize the hyperparameters for MSD data in the LAG-LLMA network. First, the number of layers is optimized. The model performing best with a 1 layer is an RMSE of 0.04 and an R^2 of 0.58. As the number of layers is increased, the performance declines dramatically, e.g., three layers result in a huge negative R^2 value of −292.20, and 10 layers result in an R^2 value of −233.14. This means that a deep transformer architecture is not required for the quite simple MSD trajectory and leads to severe overfitting or gradients being unstable. After finding the ideal number of layers, model dimension (d_model) is optimized. The best result is found at a dimension of 32 with an RMSE of 0.03 and an R^2 of 0.68. Very low dimension (e.g., 4) gave a negative value of R^2 and overly high dimension (e.g., 1024) also failed (R^2 of −69.10), which could imply that a middle-range feature space is needed for learning MSD trend. Next, the number of attention heads (n_heads) is optimized. One head of attention gives the maximum R^2 of 0.76, while increasing the number of heads to 16 decreases the R^2 to 0.64. This indicates that adding the multiparcel mechanism does not befit in providing extra benefit regarding this particular univariate time series data. After that, the dropout rate is optimized to avoid over fitting. A dropout rate of 0.05 shows the best balance in terms of R^2, which reaches 0.88 with an RMSE of 0.02. Following this, the learning rate is optimized. The model is extremely sensitive to this parameter as the best performance is observed for 7.00×10^{-4} (R^2 = 0.90). Deviating strongly from this value, including 1.00×10^{-6} and 1.00×10^{-2}, leads to large negative R^2 values. The batch size is then optimized, and it turns out that a batch size of 32 is optimal (R^2 = 0.90). Larger batch sizes have negative value of R^2. Different optimizers are then tried out with AdamW yielding the best accuracy (R^2 = 0.93) compared to Adam, RMSprop or SGD. Finally, the percentage for the data to train is optimized. The LAG-LLAMA framework shows its best performance when using 70% of the data, where the R^2 and RMSE are 0.98 and 0.01, respectively. This shows that, in contrast to its poor performance using Hbond as a data source, the LAG-LLAMA model can be used to accurately predict MSD behavior if adequate training data and a series of optimized hyperparameters are available.

3.1.1.10. Comparison of ESN, GRU, LSTM, and LAG-LLAMA Architectures for Data for MSD. This study compares ESN, GRU, LSTM, and LAG-LLAMA frameworks in predicting MSD data. The ESN framework is the most efficient model, as it has an excellent R^2 value of 0.99 and an RMSE of 1.58×10^{-3} using only 10% of the data and training it. This therefore shows that the reservoir computing architecture is uniquely suited to capture the linear and superdiffusive trends in the MSD trajectory with minimal computational cost and data requirement. In comparison, the LSTM framework also shows high accuracy in MSD data, unlike the Hbond data. The LSTM model attains an R^2 of 0.99 and an RMSE of 0.01 but required 40% of the data to get there. It is also less data efficient than the ESN. Similarly, the GRU framework results in a good performance with a R^2 score of 0.95 and an RMSE of 0.02; however, the GRU model requires 70% of data to learn the result and is unstable with a deep layer configuration. The transformer-based LAG-LLAMA model also proves to be able to model MSD dynamics with an R^2 of 0.98 and an RMSE of 0.01. However, just like the GRU, it requires a large amount of data (with 70% of the data) and a lot of hyperparameters tuning to avoid overfitting in the model, especially when it comes to layer length and dropout levels. Consequently, while all four models can be used to predict MSD behavior when optimized, the ESN is still the superior option considering that it can achieve maximum accuracy (R^2 = 0.99) with significantly less data (10%) compared to LSTM (40%), GRU (70%), and LAG-LLMA (70%).

3.2. AI-Generated Results

3.2.1. Stability of Hydrogel Framework. Figure 4 shows the Hbond dynamics between the moieties, which are analyzed using the ESN framework. After the production run, a total of 2000 frames of Hbond data are collected and prepared for training and testing. Of these, 10% of the data, corresponding to 200 frames, is used for training the ESN framework. The choice of using a small fraction of the data for training allows the model to demonstrate its efficiency in learning complex temporal patterns without requiring large data sets. During training, the ESN achieves a very high performance with an R^2 value of 0.99 and a low RMSE of 9.03×10^{-3}. These results indicate that the ESN is able to capture the underlying nonlinear dynamics of the Hbond interactions with high accuracy, while keeping the prediction error minimal. After training, the remaining 90% of the data, corresponding to 1800 frames, is used to test the model. This large testing data set ensures a reliable evaluation of the ESN's generalization ability. The testing results also show excellent performance, with an R^2 value of 0.99 and an RMSE value of 10.58×10^{-3}. The similarity between the training and testing values confirms that the ESN does not overfit the training data and is capable of generalizing the learned dynamics to unseen data. This robust performance implies that the ESN framework is sufficiently generalizable to model the nonlinear Hbonding dynamics of other chemically cross-linked hydrogels, such as poly(vinyl alcohol) or poly(ethylene glycol), thereby serving as a versatile tool for evaluating structural stability across different soft material.

Figure 5a further supports these findings by showing the percentage error of the predictions within ±0.75% of the Hbond data. This very small error range signifies that the ESN predictions closely match the actual Hbond values across the entire data set, with only minimal deviations. Such consistency indicates that the model not only learns the underlying dynamics but also maintains stability over long time series without accumulating large errors. This level of accuracy is particularly important in capturing molecular interactions, where even small deviations can affect the interpretation of structural and dynamical properties. The narrow error band therefore

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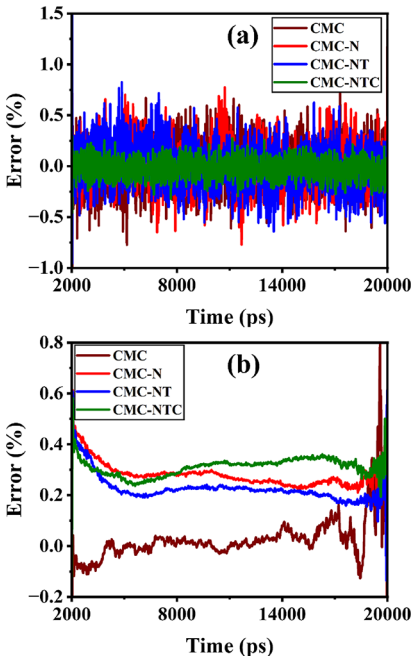


Figure 5. Percentage errors for ESN network (a) Hbond and (b) MSD.

highlights the reliability of the ESN in reproducing the fine-scale fluctuations of Hbond dynamics.

Figure 4a–d shows the hydrogen bonds between the polymer and the polymer framework. The polymer framework consists of functional groups such as hydroxyl, ether, and carboxylic moieties, which act as Hbond donors and acceptors. The average number of H-bonds is tabulated in Table 2. The order of

Table 2. Average Number of Hbonds

polymer name	average no. of Hbonds between polymer and polymer	average no. of Hbonds between polymer and solvents	percent of Hbonds forming in polymer chain to the polymer chain	percent of Hbonding forming in polymer chain to solvents
CMC	68	152	30.9	69.1
CMC-N	71	150	32.2	67.8
CMC-NT	69	155	30.9	69.1
CMC-NTC	299	521	36.5	63.5

the Hbond is CMC < CMC-NT < CMC-N < CMC-NTC. The lowest number of Hbonds shows in the CMC framework because it has the lowest number of Hbond acceptors and donors compared to the other polymer frameworks studied in this work. After that, the CMC-N polymer exhibits the second-lowest polymer–polymer Hbonding. This phenomenon arises due to the carboxylic functionalities within the norbornene polymer's structure, facilitating additional Hbonding interactions between polymer chains. Then, the CMC-NT polymer shows the third lowest Hbond acceptor and donor between the polymer and the polymer networks. This is because the dithiothreitol has a couple of hydroxyl moieties and the thiol moieties, which can add extra Hbonds between the networks. But, the CMC-NT network shows fewer Hbond than the CMC-N polymer, which signifies that it forms more Hbond with the solvent molecules compares to the polymer networks. The CMC-NTC network shows a maximum average Hbond within the polymer frameworks. The CMC-NTC polymer chain shows 4.2 folds of Hbond compared to the CMC-N polymer chains. This is because the cross-links take the networks closer. As a result, the Hbonds between the networks increase significantly which also impacts on the mechanical property and stability of the polymer frameworks.

Figure 4e–h shows the Hbond between polymer networks and the solvent networks. The order of Hbond is as follows: CMC-N < CMC < CMC-NT < CMC-NTC, with average numbers of Hbond being 150, 152, 155, and 521, respectively. On average, 63.5 to 69.1% of the Hbonds exist between the polymer and solvents, indicating water molecule penetration through the gel networks, which is confirmed in Section 2.2. The CMC-N network displays the lowest Hbond between polymer and solvent networks. Norbornene is a hydrophobic molecule. After cross-linking of norbornene into CMC, it enhances the hydrophobicity of the polymer framework. As a result, it forms a greater number of Hbond polymer frameworks compared to the water molecules. The second highest Hbonds show between CMC-NT and water molecules. The dithiothreitol is a hydrophilic material that enhances the hydrophilicity of the CMC-NT network. The hydrophilicity of the dithiothreitol has been increased in the presence of extra hydroxyl and thiol moieties, which leads to an increase in the number of Hbonds formed with water. This also enhances the hydrophilicy of CMC-NT frameworks compares to the CMC frameworks. The most significant increase in Hbonding is observed in the CMC-NTC network. This network contains a higher number of Hbond acceptor and donor moieties due to the presence of multiple number of the CMC-N and the CMC-NT chains. The CMC-NTC chains self-assemble and form an enclosed network, which can further reduce the Hbonding with solvent molecules, as shown in Table 2. This result suggests that the CMC-NTC polymer has higher mechanical strength, as suggested by Figure 4d. However, the Hbond between polymer and water suggests that it forms a soft gel-type material.

3.2.2. Mean-Squared Displacement. Figures 6 and S16–S20 of the Supporting Information show the MSD of the different moieties, which are analyzed using the ESN framework. After the production run, a total of 2000 frames of MSD data are collected. From these, 10% of the data, corresponding to 200 frames, were used to train the ESN framework. The training results show excellent accuracy, with an R^2 value of 0.99 and a very low RMSE of 1.02×10^{-3}. These values indicate that the ESN successfully captures the temporal evolution of the MSD during the training stage and can reproduce the general dynamics with very small deviations. The remaining 90% of the data, corresponding to 1800 frames, is used to test the model. Testing results also confirm high performance, with an R^2 of 0.99 and an RMSE of 1.58×10^{-3}. The close agreement between the training and testing performance demonstrates that the ESN generalizes well and avoids overfitting. This highlights the ability of the ESN to reliably model the diffusion behavior of moieties using only a limited portion of the data for training. Consequently, this data-efficient approach establishes a foundation for applying reservoir computing to a wider range of hydrogel systems, enabling the rapid prediction of long-term transport phenomena in complex macromolecular networks without relying on computationally prohibitive production runs.

Figure 6 further provides insight into the rate of percentage errors for MSD predictions. At the beginning of the trajectory, the error percentage is higher. This occurs because the MSD data are nearly linear in the early stage, where statistical averaging across many short-time displacements dominates. In this region, the ESN finds it more challenging to distinguish the

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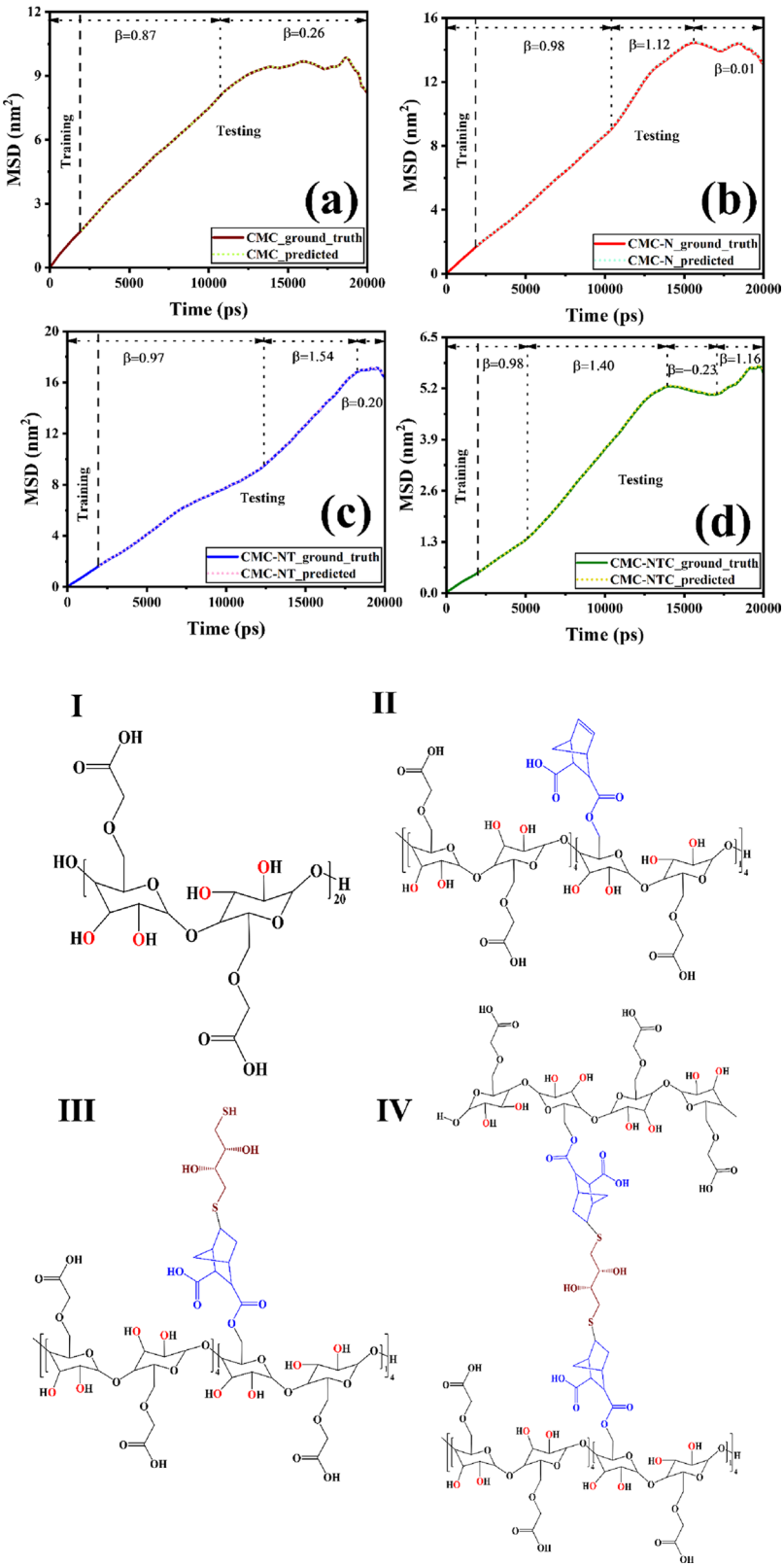


Figure 6. MSD curve of oxygen atoms of hydroxyl moieties in the polymer backbone. (a) Illustration of the MSD curve of CMC, (b) illustration of the MSD curve of CMC-N, (c) illustration of the MSD curve of CMC-NT polymer chain, and (d) representation of the MSD curve of CMC-NTC polymer chain, I. schematic diagram of CMC, II. schematic diagram of CMC- N, III. schematic diagram of CMC-NT, and IV. schematic diagram of CMC-NTC.

signal from noise, leading to slightly higher prediction errors. As time progresses, the MSD data begin to show nonlinear characteristics. The ESN captures these nonlinear patterns effectively, and as a result, the error percentage stabilizes and becomes nearly constant over most of the trajectory. Toward the end of the MSD data, the statistical averaging decreases due to

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fewer available time points, which increases fluctuations and noise. Consequently, the ESN also exhibits a higher percentage error in this region.

Figure 6 and S16−S18 illustrate the MSD profiles of hydroxyl, carboxyl, and ether groups present in the polymer framework backbone. Among these, the CMC-NT polymer exhibits the highest MSD value. Due to the dithiothreitol functionalization of the CMC-NT framework, it enhances the hydrophilicity of the polymer framework. As a result, it easily moves in the aqueous media. The CMC-NT framework shows three distinct regions in the MSD curve. Initially, the MSD exhibits linear characteristics with a time span of 12.3 ns. The logarithmic slope of the MSD is less than one, which suggests the subdiffusive behavior. The subdiffusive behavior is observed due to the interactions between the hydroxyl moiety and water molecules and the intramolecular and intermolecular Hbonding between the molecules. This Hbonding creates a cage which hindered the molecular motion. After 12.3 ns, there is an acceleration in the MSD curve until 18.3 ns. Here, the MSD profile shows the superdiffusive behaviors, which are suggested by the logarithmic slope value. This superdiffusive behavior is observed due to the molecular hopping mechanism. This occurs in solid-like liquids, where molecules move by hopping between positions due to loose packing.[68] After that, the MSD curve shows a plateau until the end of the dynamics. Due to the lower statistical averaging in the MSD data. The second highest MSD value is observed in the CMC-N polymer chain. The MSD profile initially behaves linearly for 10.4 ns. In this interval, the logarithmic slope of the MSD value is less than unity, which signifies the subdiffusive behavior of the hydroxyl moiety. This is due to the H-bonding between the polymer to solvents and the polymer to the polymer. After that, there is an acceleration in the MSD curve until 15.9 ns. In these time intervals, the observation indicates hydroxyl moieties exhibit superdiffusive characteristics, as the logarithmic slope of the MSD is greater than one. This superdiffusion is observed due to the molecular hopping mechanism. After 15.9 ns, the MSD curve shows the nonlinear behavior until the end of the dynamics, which is due to the less statistical averaging of data. The third-highest MSD value is shown in the CMC curve. Initially, the linear behavior is observed in the MSD profile for 10.8 ns. The observation of MSD slopes indicates the subdiffusive behavior of the hydroxyl moieties, as suggested by a logarithmic slope value below unity. After that, the curve shows the nonlinear behavior until the end of the dynamics. In the current period, the moieties exhibit anomalous diffusion, indicating the presence of intricate dynamics or underlying mechanisms that deviate from conventional diffusion processes. The lowest MSD value is shown in the CMC-NTC polymer framework. In the initial 5.1 ns, linear behavior is characterized by the MSD profile, with a logarithmic slope below unity, indicating a subdiffusive behavior of the particle. This subdiffusive behavior is observed due to Hbonding. After that, there is an increase in the MSD profile until 13.9 ns. In these time intervals, the logarithmic slope of MSD exceeds one, indicating the superdiffusive nature of the hydroxyl moieties, which is due to the molecular hopping mechanism. After 13.9 ns, the MSD profile exhibits a concave curvature throughout the dynamics. This curvature is observed due to anomalous diffusion. Figures S19 and S20 shows the MSD of the unsaturated hydrocarbon and thiol moieties present in the system. The MSD curve shows the subdiffusive and superdiffusive behavior throughout the dynamics. The subdiffusive behavior is observed due to the Hbonding, and the superdiffusive behavior is observed due to the molecular hopping mechanism.

3.3. MD-Generated Data Analysis

3.3.1. Swelling Ratio. Figure 7 illustrates the swelling ratios of the hydrogel frameworks, displaying the order as CMC-N < CMC < CMC-NT < CMC-NTC, with respective mean values of 175.81, 231.04, 293.28, and 362.20. The CMC-N network shows a lower swelling ratio due to the smaller number of nonfreezing waters due to the polymer chain's interaction, as suggested by the Figure 4f. Due to the smaller amount of nonfreezing water, it also attracts a smaller amount of freezing water, which is the reason for the lowest swelling ratio. The CMC-N polymer framework shows a lower swelling ratio compared to the CMC network. This is due to norbornene functionalization, which enhances hydrophobicity of the CMC-N frameworks. The CMC-NT framework shows the second highest swelling ratio due to dithiothreitol functionalization. The dithiothreitol moiety has a couple of hydroxyl groups and thiol groups, which enhance nonfrezing and freezing water. The CMC-NTC network shows the highest swelling ratio, with a mean value of 362.20 and a standard deviation of 13.51. The CMC-NTC framework attracts a high amount of the nonfrezing water, which is suggested by Figure 4h. As it has a high amount of nonfreezing water, which also attracts a high amount of freezing water, enhancing the swelling ratio of the polymer frameworks. This cross-linked hydrogel forms a more open and stable network that can trap more water molecules. It also shows that thiol−ene click chemistry enhances the structure and water absorption of the hydrogel.

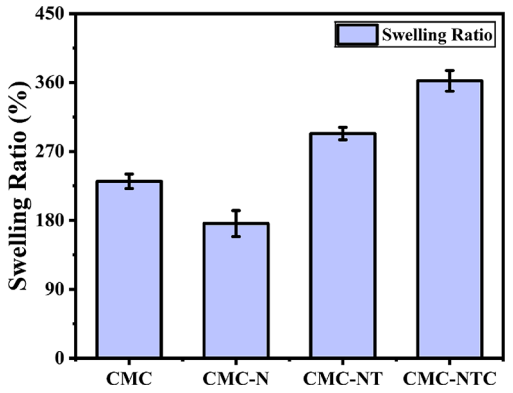


Figure 7. Swelling ratio of hydrogels.

3.3.2. Spatial Density Distribution. Figures 8 and S21 show the 2D number density distribution for polymer moieties and water molecules, respectively. The CMC density map displays a nearly round shape with a high number density of 57.80 nm^{-3} at the center, which gradually decreases to 11.56 nm^{-3} toward the surface. This suggests that the hydrophobic part of the CMC polymer chains is densely packed at the core and the hydrophilic part of the CMC becomes less concentrated at the edges due to absorption of water. In contrast, the water distribution map indicates that water molecules are present throughout the CMC hydrogel. At the center, the water density is 62 nm^{-3}, while at the outer surface or periphery, it increases to 120 nm^{-3}. This pattern shows that water not only penetrates the hydrogel but also accumulates more at the boundary. The uniform water distribution at the center of the CMC implies that the hydrogel structure allows water to enter and spread evenly inside, which is a key feature of a swollen and hydrated network.

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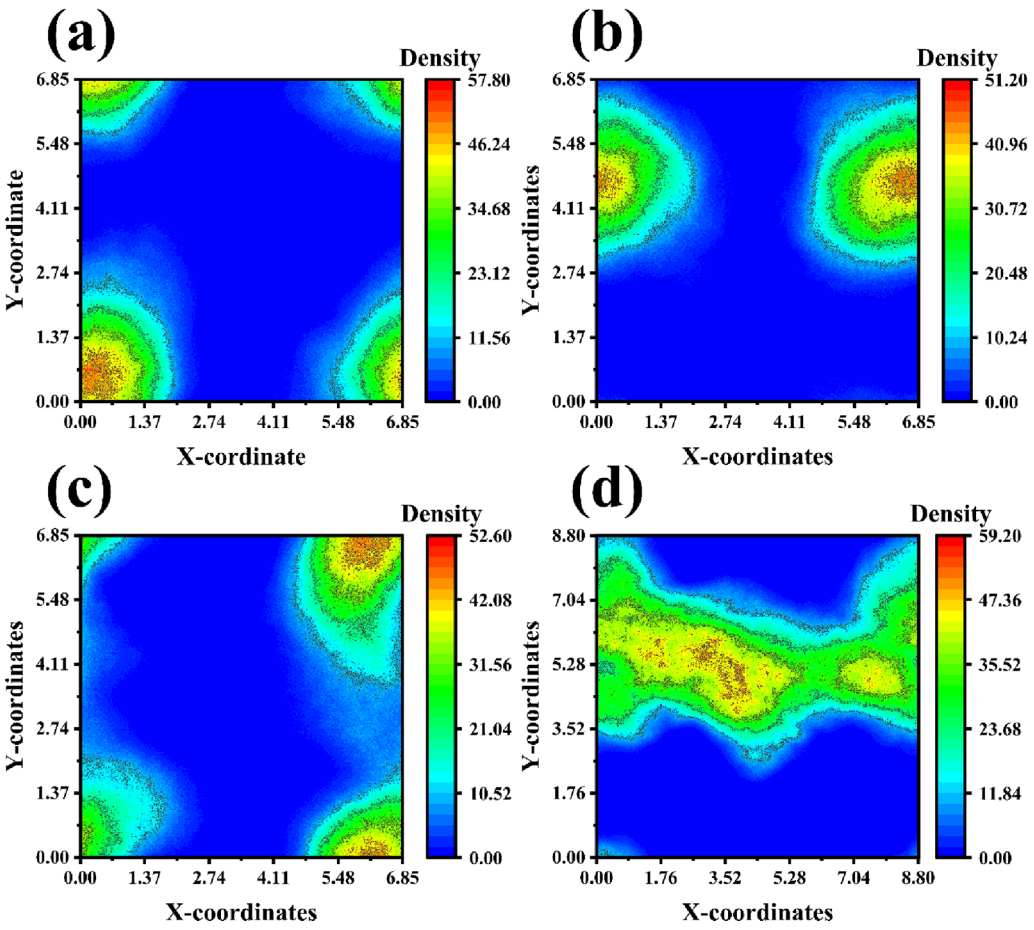
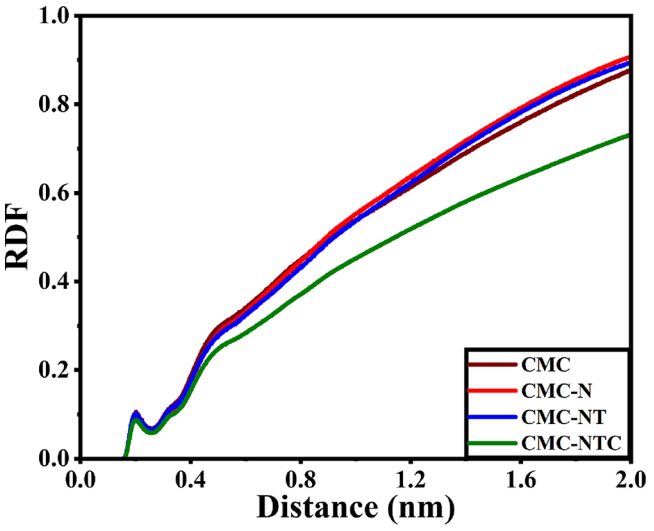


Figure 8. 2D number density counter plot (a) CMC, (b) CMC-N, (c) CMC-NT, and (d) CMC-NTC.

The CMC-N core shows a density of 51.20 nm^{-3}, which gradually decreases toward the surface of the CMC-N framework. However, the distribution of water molecules shows a reverse order. The CMC-N framework is self-assembled in such a way that the hydrophobic part is presented in the core of the polymer matrix, and the hydrophilic part is on the surface. As a surface water number density is higher, and in the core, it is lower. The CMC-NT core shows a density of 52.60 nm^{-3}, which gradually decreases toward the periphery, indicating a dense polymer network at the center and a looser structure at the edges. This suggests that the hydrophobic structure of the CMC-NT framework is concentrated in the core and the hydrophilic part of the CMC-NT framework is present on the surface, as also indicated by the water distribution in Figure S21b. The water distribution suggests that the water molecules are penetrating the CMC-NT framework. Compared to the CMC distribution, it shows a lower value at the core. This suggests that the dithiothreitol functionalization enhances the hydrophilicity of the CMC network. This enhances the gelling property of the CMC network. The CMC-NTC reveals a higher polymer density at the core of 59.20, which gradually decreases toward the periphery. However, the water density shows the opposite trend, being lower at the core (64.80) and significantly higher at the periphery (152.80). This inverse relationship signifies that the CMC-NTC polymer network forms a dense, structured core that restricts water penetration, likely due to inter- and intra-Hbonding and the partial hydrophobic character introduced by the norbornene functionalities. The hydrophilic backbone of CMC attracts some water, but the cross-linked network's hydrophobic regions limit water access to the core, forcing water molecules to accumulate more at the periphery. However, compared to the other structure, it has higher density in the core, which suggests the enhanced mechanical property of the CMC-NTC framework. However, it absorbs a sufficient amount of water, which suggests the formation of a gel network. However, the neighborhood of the CMC, CMC-N, and CMC-NT surface shows that the nonfreezing water tries to disintegrate the hydrogel structure. But the CMC-NTC does not show this type of structural disintegration. Due to the thiol-norbornene cross-linking in the CMC-NTC framework, it maintains the structural integrity.

3.3.3. Radial Distribution Function. Figures 9 and S22–S24 show the RDF curve between the ether groups, hydroxyl moiety of the carboxylic group, oxygen of the hydroxyl group, and oxygen of the carbonyl group of polymer backbone (shown in red color) to the hydrogen of the water. The first solvation shell is observed at 0.25 nm with CN values of 0.25, 0.23, 0.24, and 0.20 for CMC, CMC-N, CMC-NT, and CMC-NTC, respectively. The first solvation shell is observed in the Hbonding region. The order of the CN is CMC > CMC-NT > CMC-N > CMC-NTC. The CN number suggests that the CMC and CMC-NT polymers attract more water molecules than other polymers, which is also proof of the Hbonding occurrence. Due to the cross-linking, the CMC-NTC polymer shows the lowest CN number of the other polymer networks. The RDF value is less than unity at a far distance, which signifies the formation of the cluster. The other moiety of the CMC backbone hydroxyl and carbonyl group shows the first solvation

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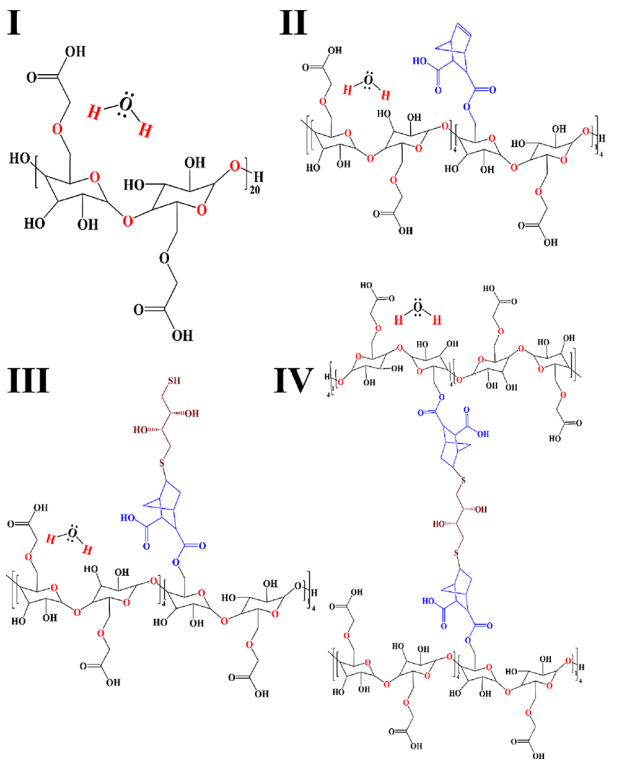


Figure 9. Schematic representation of the radial distribution function of oxygen atoms in the ether group of the polymer backbone to water hydrogen atoms. I. Schematic representation of CMC interacting with water, II. CMC-N interacting with water, III. CMC-NT interacting with water, and IV. Schematic representation of CMC-NTC interacting with water.

shell in the Hbonding region, except the hydroxyl in the carboxylic group which is observed in the nonbonded van der Waals (vdW) interactions regions. This suggests that hydroxyl, carbonyl, and ether groups are contributing to the formation of the Hbond between the polymer water molecules. However, the clear first solvation shell is observed in the ether moiety of the CMC backbone. The ether oxygen in CMC is a weak Hbond acceptor compared to hydroxyl and carboxyl groups. It does not participate strongly in hydrogen bonding with water but can still organize water molecules around itself due to steric and electronic effects, leading to a well-defined first solvation shell. However, both hydroxyl and carboxyl groups are strong Hbond donors and acceptors, enabling them to form multiple, dynamic H-bonds with surrounding water molecules, which leads to a more diffuse and less structured solvation shell. Additionally, the CMC forms intramolecular hydrogen bonding, which particularly involves carboxyl and hydroxyl groups. This intermolecular Hbond competes with water for Hbonding sites and thereby reduces the number of available sites for water to form a well-defined solvation shell. Figure S25 shows the RDF between the unsaturated hydrocarbon present in the CMC-N and CMC-NTC framework and the oxygen of water. The first solvation shell observed in the vdW interaction regions, which suggests the weak interactions with water. This further signifies the hydrophobicity of the networks. Figure S26 thiol moiety presents in the CMC-NT and CMC-NTC polymer frameworks to the hydrogen of the water. The first solvation shell is observed in the vdW interaction regions, which suggests the weak interactions with water.

Figure 10 shows the RDF between unreacted thiol and unsaturated hydrocarbon moieties within the CMC-NTC polymer chain (shown in red color). The first solvation shell is observed at 0.70 nm with a CN value of 0.23. The first solvation shell comes due to the vdW nonbonded interactions. This suggests a very weak and infrequent spatial association between

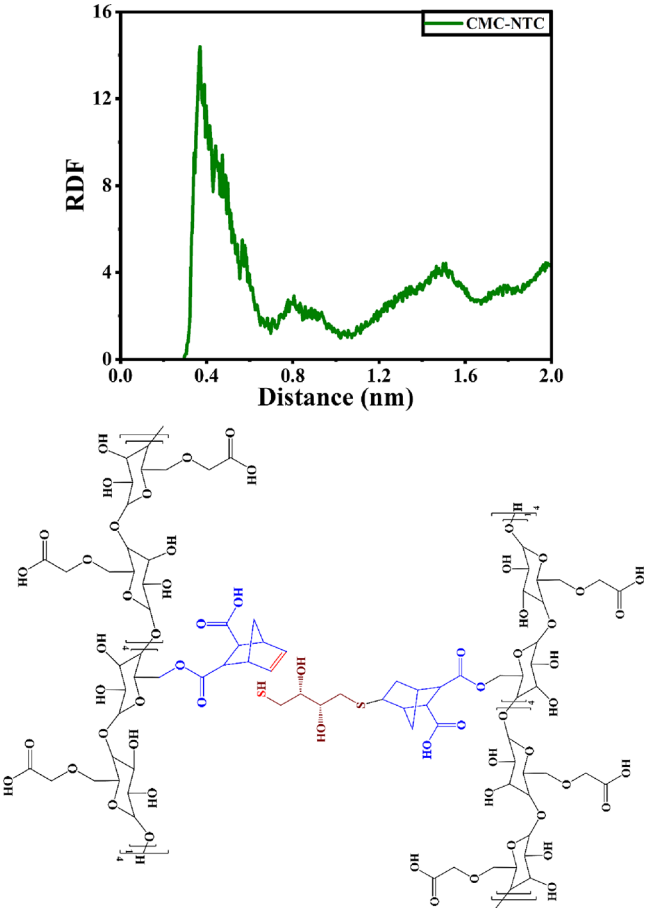


Figure 10. Schematic representation of the radial distribution function of unreacted thiol moieties interacting with unreacted unsaturated hydrocarbon moieties present in the CMC-NTC and schematic representation of thiol moieties interacting with unsaturated hydrocarbons in the CMC-NTC.

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these groups. The low *CN* means, on average, less than one unsaturated hydrocarbon is found within the first shell of a thiol group, suggesting minimal direct interaction or clustering. This spatial arrangement reflects the heterogeneous and loosely packed nature of the cross-linked network.

4. CONCLUSIONS

The work provides a unified approach that integrates artificial intelligence and atomistic MD simulations to determine how chemically cross-linked CMC hydrogels change with time. Thiol−ene cross-linked network has increased structural stability and swelling capacity in comparison to non-cross-linked ones. The structural and transport properties are trained in the AI framework. The comparison demonstrates that ESN in comparison to industry standard deep learning models; GRU, LSTM, and transformer-based LAG-LLAMA networks are unable to learn the nonlinear Hbond variations, but the ESN can do this with 10% of the trajectory information. Although deep learning models can capture the smoothness of super-diffusive trends of the MSD on the nonoptimized model, the models require 40−70% of the training data to reach the peaks compared to the ESN that maintains purely a high level of data efficiency and stability.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcb.5c08098.

Detailed information on simulation systems, initial geometry of polymers, dynamic behavior of hydrogels at different time intervals, and the ESN hyperparameters (PDF)

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Notes

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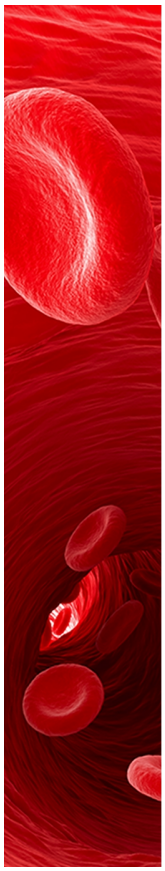
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